

Evaluation of small-angle scattering data of charged particles using the generalized indirect Fourier transformation technique

G. Fritz, A. Bergmann, and O. Glatter

Citation: *J. Chem. Phys.* **113**, 9733 (2000); doi: 10.1063/1.1321770

View online: <http://dx.doi.org/10.1063/1.1321770>

View Table of Contents: <http://jcp.aip.org/resource/1/JCPSA6/v113/i21>

Published by the [American Institute of Physics](#).

Additional information on J. Chem. Phys.

Journal Homepage: <http://jcp.aip.org/>

Journal Information: http://jcp.aip.org/about/about_the_journal

Top downloads: http://jcp.aip.org/features/most_downloaded

Information for Authors: <http://jcp.aip.org/authors>

ADVERTISEMENT



Goodfellow
metals • ceramics • polymers • composites
70,000 products
450 different materials
small quantities fast
www.goodfellowusa.com

Evaluation of small-angle scattering data of charged particles using the generalized indirect Fourier transformation technique

G. Fritz, A. Bergmann, and O. Glatter^{a)}

Institut für Chemie, Karl-Franzens Universität Graz, Heinrichstrasse 28, A-8010 Graz, Austria

(Received 21 June 2000; accepted 11 September 2000)

Small-angle x-ray and neutron scattering are widely used techniques to study the structure of colloidal particles in the size range up to 100 nm. The indirect Fourier transformation technique is well established to obtain model free real space information, but the interpretation of the results is limited to cases where particle interaction can be neglected. The extended generalized indirect Fourier transform (GIFT) allows one to separate inter- and intraparticle effects, but needs models for the particle interaction. We present the application of three different models for the calculation of interaction effects of charged particles, represented by the structure factor. With this extension, useful real space information can be obtained by the GIFT method for solutions with volume fractions up to about 0.3 without any assumption for the shape of the particles. Only the interaction effects need a model assumption, and the parameters determined from this model can give some additional information. Simulations show that it is impossible to determine charge and ionic strength simultaneously. There exists another ambiguity between the parameter sets for charge, radius, and volume fraction, but we show how this problem can be overcome in most cases. The practical applicability of the method is demonstrated by means of the micellar system CTAB in different concentrations from 1% up to 20% and with varying amounts of added salt to screen the charges and change the particle shape. © 2000 American Institute of Physics. [S0021-9606(00)02145-0]

I. INTRODUCTION

Colloidal particles can be studied by means of small-angle x-ray and neutron scattering (SAXS and SANS) to reveal their size, shape, and internal structure.^{1–5} But the resulting intensities $I(q)$ at different scattering vectors $\mathbf{q}$ with

$$|\mathbf{q}| = \frac{4\pi}{\lambda} \sin \theta, \quad (1)$$

depend not only on these parameters. Neighboring particles can influence the scattering curve considerably, leading to interparticle effects represented by the so-called structure factor $S(q)$, while the intraparticle effects are described by the form factor $P(q)$.^{6,7} The influence of the structure factor on the experimental results depends on the preferred ordering of the particles, which is obviously a result of the interaction between the particles. As a consequence the structure factor can be neglected for many dilute uncharged samples due to the large mean distance between the particles. Charged colloids interact strongly leading to the breakdown of this assumption even at low concentrations unless the potential is screened to a high extent.

In the case of monodisperse, homogeneous, globular particles, the total scattering intensity can be expressed by

$$I(q) = NS(q)P(q) \quad (2)$$

with the number of particles N . This factorization allows a separation of inter- and intraparticle effects by means of the generalized indirect Fourier transformation (GIFT) technique,⁸ which is an extension of the indirect Fourier transformation IFT.⁹ The IFT technique transforms the scattering curve into real space resulting in a pair distance distribution function (PDDF) including corrections for instrumental broadening. This transformation is model free and there are no constraints except the maximum dimension of the particle. Calculation is fast and stable, but the lack of separation of form factor and structure factor leads to PDDFs that are hardly interpretable for strongly interacting systems.

The GIFT technique uses the same model free approach for the PDDF, which corresponds in this case only to the form factor. At the same time, a model for the structure factor is assumed and fitted to the data according to Eq. (2). This simultaneous approximation of form and structure factor is only possible due to the different mathematical properties of the two functions. Nevertheless, the problem is now highly nonlinear and has to be solved in an iterative way.¹⁰

As stated previously, a model for the structure factor has to be assumed for use in the GIFT technique. Therefore models for charged particles are necessary to extend the method to this important group of colloids. Several ways to calculate the structure factors of charged particles have been introduced, using different closure relations for the Ornstein–Zernike equation.¹¹ Hayter and Penfold presented an analytical solution for this structure factor in the mean spherical approximation (MSA),¹² which was subsequently modified to the rescaled mean spherical approximation (RMSA) to

^{a)}Author to whom correspondence should be addressed; electronic mail: otto.glatter@kfunigraz.ac.at

rule out pair distribution functions with unphysical negative values near contact.¹³ Another very effective and widely used closure relation is the hypernetted-chain (HNC) approximation. Strongly correlated systems deviate systematically from the HNC and the widely used Percus–Yevick¹⁴ closure relations, which led Rogers and Young (RY) to propose a mixture of these two schemes,¹⁵ the RY closure relation. Potentials, which also have an attractive part, can be described thermodynamically self-consistently by mixing HNC with the so-called “soft MSA,”^{16,17} closure, resulting in the HMSA closure relation.¹⁸ There also exist a variety of structure factor models for polydisperse charged spheres.^{19–21}

The above-mentioned closure relations vary in the description of different potentials: The Percus–Yevick closure relation works particularly well for short-range interactions while the HNC is best for systems with long-range interactions. With the mixture of these two schemes, the RY closure relation is designed to describe strongly correlated systems of repulsively interacting particles. The simple MSA is only applicable for weakly interacting charged particles, while the RMSA scheme suffers from its underestimation of the structure. Nevertheless it can be used for a simple and fast fit to the data. The “soft MSA” can be used if the potential can be separated into a short-range repulsion and a weak long-range contribution. The HMSA scheme can treat potentials that have an attractive part.

These limitations of the closure relations due to the interaction potentials led us to restrict ourselves to RMSA, HNC, and RY to evaluate the scattering functions of charged and thereby strongly and repulsively interacting particles.

We want to demonstrate the possibilities and limitations of the use of the structure factors derived from the different closure relations in the GIFT method for simulated and measured scattering data. The experimental data are for micelles of the cationic surfactant CTAB. This substance has been selected, not because of its novelty, but because it has already been characterized well.^{22–26} Therefore it allows one to compare the results obtained by the GIFT technique with the parameters known from the literature.

II. THEORY

The GIFT method approximates the form factor, represented by its PDDF $p(r)$ by a linear combination of a finite number (n) of cubic B-spline functions $\varphi_v(r)$ in real space:

$$p(r) = \sum_{v=1}^n c_v \varphi_v(r). \quad (3)$$

The expansion coefficients c_v are the unknown parameters. The PDDF is connected to the particle form factor $P(q)$ by the spatially averaged Fourier transformation

$$P(q) = 4\pi \int_0^\infty p(r) \frac{\sin(qr)}{qr} dr. \quad (4)$$

If the particle interactions and the structure factor were known, the expansion coefficients could be determined easily using the fact that the Fourier transformation and possible

smearing effects are linear transformations. As a result, the Fourier-transformed and smeared splines $\psi_v(q)$ represent the scattering curve according to

$$I(q) = P(q)S(q, \mathbf{d}) = \sum_{v=1}^n \{c_v \psi_v(q)\} S(q, \mathbf{d}), \quad (5)$$

where the expansion coefficients c_v are the same as in Eq. (3) and the parameters $\mathbf{d}$ determine the structure factor. A stability criterion has to be taken into account to get the correct results for the coefficients c_v .⁹ On the other hand, the parameters $\mathbf{d}$ are usually also unknown, giving rise to a highly nonlinear problem. Such problems can no longer be solved by standard linear least-squares algorithms, but have to be treated by iterative procedures until the parameters c_v and $\mathbf{d}$ lead to a global minimum of the mean deviation. The Boltzmann simplex simulated annealing (BSSA) algorithm has proved to be very reliable and fast to find the global minimum.¹⁰

The correct structure factor is a crucial point to calculate the PDDF by means of the GIFT technique. Its calculation is based on the solution of the Ornstein–Zernike equation¹¹

$$h(r) = c(r) + n \int d\mathbf{r}' c(\mathbf{r}-\mathbf{r}') h(\mathbf{r}'), \quad (6)$$

where the total correlation function $h(r)$ is related to the direct correlation function between the particles $c(r)$ and the indirect contributions, which are represented by the integral. But the direct correlation function is unknown, and further relations, the so-called “closure relations,” are needed as approximations to solve the integral equation.

The first of the closure relations used for the structure factors examined in this paper is the mean spherical approximation, which leads to the direct correlation function

$$c(r) = -\beta v(r), \quad (7)$$

while $h(r) = -1$ is assured for $r < \sigma$, with σ as diameter of the particles, to assure hard impenetrable spheres. The particles potential is introduced by $v(r)$ and $\beta = 1/(k_B T)$ is an energy term. As already mentioned, the pure MSA in combination with a Coulomb potential can give rise to unphysical results near the contact distance of two spheres. This problem can be avoided by an iterative rescaling procedure, which increases the apparent radius of the particle until the physically correct solution is achieved.¹³

Another possibility to provide an additional relation between the functions appearing in the Ornstein–Zernike equation is the hypernetted-chain closure relation (HNC)

$$g(r) = e^{-\beta v(r) + h(r) - c(r)} \quad (8)$$

with the pair distribution function $g(r) = h(r) + 1$.

The empirical Rogers–Young closure relation¹⁵ is related to the HNC scheme, because it is obtained by mixing it with another, the Percus–Yevick¹⁴ approximation. The resulting equation,

$$g(r) = e^{-\beta v(r)} \left\{ 1 + \frac{1}{f(r)} [e^{f(r)(h(r)-c(r))} - 1] \right\}, \quad (9)$$

depends on the mixing function

$$f(r) = 1 - e^{-\alpha r}. \quad (10)$$

The mixing parameter α remains to be adjusted. This task can be achieved by assuring that the thermodynamic parameters, which result from the structure factor, must be self-consistent.⁷ The Yukawa (or screened Coulomb) potential $v(r)$ of a particle of diameter σ and charge Z ,

$$\beta v(r) = \frac{Z^2 e_0^2}{4\pi\epsilon\epsilon_0 kT} \cdot \frac{e^{-\kappa(r-\sigma)}}{r(1+\kappa\sigma/2)^2}, \quad (11)$$

can be introduced into these closure relations, and the solution can be calculated analytically for the MSA method or numerically for the HNC or the RY schemes. The solvent is represented by its dielectric constant ϵ and the κ is the reciprocal Debye screening length, which is determined by the ionic strength

$$\kappa = \sqrt{\frac{\beta e_0^2}{\epsilon\epsilon_0} \sum_i \rho_i}. \quad (12)$$

The sum is taken over the univalent counterions and electrolyte ions of the number densities ρ_i . Once the Ornstein–Zernike equation is solved, the structure factor can be derived for a particle density n by the following transformation:

$$S(q) = 1 + 4\pi n \int_0^\infty h(r) r^2 \frac{\sin(qr)}{qr} dr. \quad (13)$$

This structure factor contains physical constraints due to the particle potential used. These limitations to physically reasonable values for the input parameters restrict the properties of the resulting $S(q)$ function and allow its separation from the form factor.

Due to the long calculation times needed for some of the structure factors, we have limited the polydispersity to the simple model of an averaged RMSA-structure factor. But these results are highly unstable and all the data presented in this contribution are evaluated with a monodisperse approximation for the structure factor.

III. EXPERIMENT

A. Materials

Cetyltrimethylammonium bromide (CTAB MW = 364.5 g/mol) was purchased from Aldrich and was recrystallized three times in a 50:50 mixture of ethanol and acetone by the group of E. Kaler, Delaware. The solvent used for the experiments was D₂O (99.8% deuterated) from Merck. KCl (p.a.) and KBr (p.a.), were also obtained from Merck.

Solutions of the interesting molarities of KCl and KBr in D₂O were prepared and CTAB was added to these stock solutions. The concentrations are given in % (w/w) that would apply to a solution with the same molecular composition in H₂O instead of D₂O. So the percentages given in this contribution are a little bit higher than the actual concentrations in D₂O.

B. SANS

SANS experiments were conducted at the spallation source at the PSI, Switzerland. The wavelength λ used was 8 Å with a spread in wavelength $\Delta\lambda/\lambda$ of 10%. The sample-detector distances used were 2.0 and 8.0 m. The two-dimensional ³He-detector itself is made of 128×128 pixels, each with a side length of 7.5 mm. Samples were held in 2.0-mm-thick quartz cells in a temperature controlled sample holder at 37 °C. More than 2×10^6 counts were collected for every sample. The raw data were corrected for detector efficiency, scattering from the empty cell, and background radiation and radially averaged. An absolute scale of the data was obtained by H₂O calibration. All data are shown on absolute scale.

IV. SIMULATIONS

To demonstrate the applicability of the GIFT technique for charged particles, we simulated a 5% (v/v) solution of spheres of radius 2.5 nm and a charge of 25 electron charges in water at 298.15 K with 10 mM of univalent salt. The q scale was simulated from 0.1 up to 3 nm⁻¹ and smeared with a wavelength distribution $\Delta\lambda/\lambda = 0.1$. A random error $\sigma(q) = \sqrt{1.5625I(q)}$ was added to assure realistic conditions.

The simulated data are shown in Fig. 1(a) together with the approximations using various closure relations. The corresponding PDDFs are depicted in Fig. 1(b). The PDDF, which can be obtained by using the exact values used for the simulation of the structure factor, is included in this plot. This curve represents the best result that can be obtained from these data. Figure 1(c) shows the structure factors obtained from these calculations together with the theoretical structure factor which has been used for the simulation. Figure 1(d) demonstrates how the results of the GIFT technique in reciprocal space correspond to the theoretical form factor used for the simulation. An important fact that can be studied by simulations is the great ambiguity of the structure factor parameters: A wide variety of different parameter sets can yield structure factors that agree within the limits of accuracy of the evaluation program for the RMSA closure relation. The dependence of the volume fraction and charge on different radii leading to nearly the same structure factor, and hence to mean deviations which differ only by $4 \times 10^{-3}\%$, is shown in Fig. 2. The other closure relations studied, HNC and RY somehow show a similar behavior. For these cases the structure factor varies slightly, but when the radius was fixed up to 20% off the exact value of our simulation, the mean deviations of the results differ by less than 1% (results not shown).

Another criterion that may lead to important similarities between structure factors for different parameter sets is the counterbalance of charge and ionic strength. The resulting structure factors obtained for various ionic strength are shown in Fig. 3(a). The dependence of the corresponding parameters is plotted in Fig. 3(b). It must be mentioned that these calculations allowed only a variation of the charge while volume fraction and radius were kept constant. If all three parameters can be adjusted during calculation, as

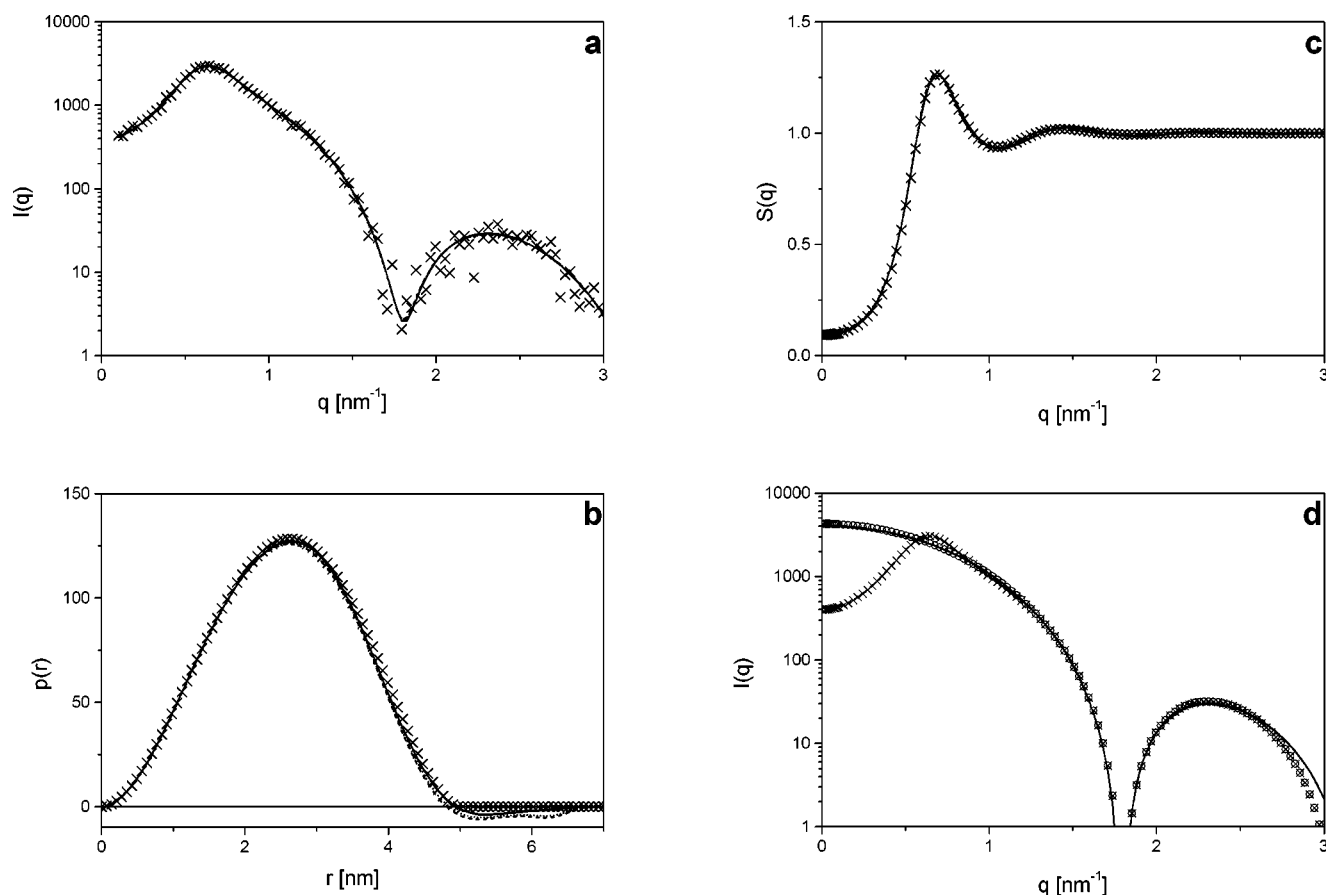


FIG. 1. Simulation of an interacting (Rogers–Young closure relation) sphere of radius 2.5, volume fraction 5%, charge 25 e , and salt concentration of 10 mM with wavelength smearing and added error ($\times$). The results using the RMSA (solid line), HNC (dashed line) and RY (dotted line) closure relations with the GIFT method agree most times within the line thickness of this plot. (a) Approximated scattering curves. (b) The corresponding pair distance distribution functions including the theoretically correct PDDF ($\times$). (c) The structure factors obtained from these calculations and the theoretical structure factor ($\times$). (d) The theoretical scattering curve ($\times$) with the resulting desmeared scattering curve (RY, dashed) and the simulated form factor ($\circ$) compared to the resulting form factor using RY (solid).

would apply for real samples, much closer similarities can be obtained for structure factors at different ionic strengths (results not shown).

V. EXPERIMENTAL RESULTS

A. Concentration dependency

Solutions of 1%, 5%, 10%, 15%, and 20% CTAB in a 10 mM KCl solution have been measured with neutron scattering (Fig. 4). All concentrations show a strong influence of repulsive interaction, which can be detected by lowered scattering intensity at low q values. This decreasing signal in the low q regime is accompanied by the formation of an interaction peak representing the higher degree of order in the sample. High CTAB concentrations also give rise to another weak shoulder at about 1.5 nm^{-1} . Evaluation by means of the GIFT technique can approximate these scattering curves reasonably well for RMSA, HNC (shown in Fig. 4), and RY. The low intensity in the low q regime and the interaction peak are due to the structure factor components (Fig. 5).

The pure form factor, after elimination of the interparticle effects, can be represented by its Fourier transform, the PDDF (Fig. 6). The shapes of these functions are up to 5%,

typical for nearly spherical micelles with a maximum dimension of about 6 nm. The addition of more CTAB leads to an elongation of the particles. The diameter of these particles is reduced to about 5 nm while the length increases at least up to 10 nm.

The form factor $P(q)$ can be used at $q=0$ to calculate the aggregation number and in consequence the parameter f , which is the apparent fraction of charged molecules in the micelle (Table I).

B. Salt addition

The addition of KCl changes the interaction potential by screening the charged macroions. This change in the interaction leads to essential variations in the structure factors obtained for the samples (results not shown). Nevertheless the resulting PDDFs remain nearly the same (Fig. 7). Only the relatively high concentration of 80 mM KCl shows significant deviations. Like in the CTAB concentration series, an elongation of the micelles can be observed, but it is far less pronounced.

KBr addition gives rise to an obvious change in micelle shape (Fig. 8). CTAB in 10 mM KBr still looks very similar to the nearly spherical micelles found in KCl solutions. Nev-

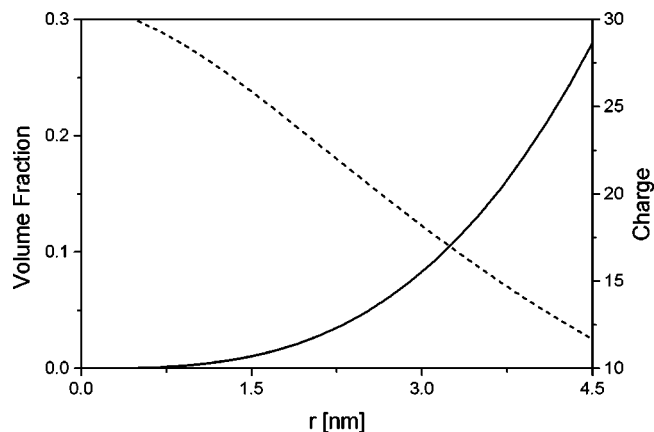


FIG. 2. Simulation like in Fig. 1, variation of the interaction radius: Combinations of charge (dashed line) and volume fraction (solid line), that lead to the same structure factor for the RMSA closure relation like in Fig. 1(c).

ertheless 30 mM KBr are sufficient to elongate the micelles considerably with a maximum dimension of at least 12 nm. A further increase of KBr concentration makes this dramatic change more evident. The diameter of these elongated mi-

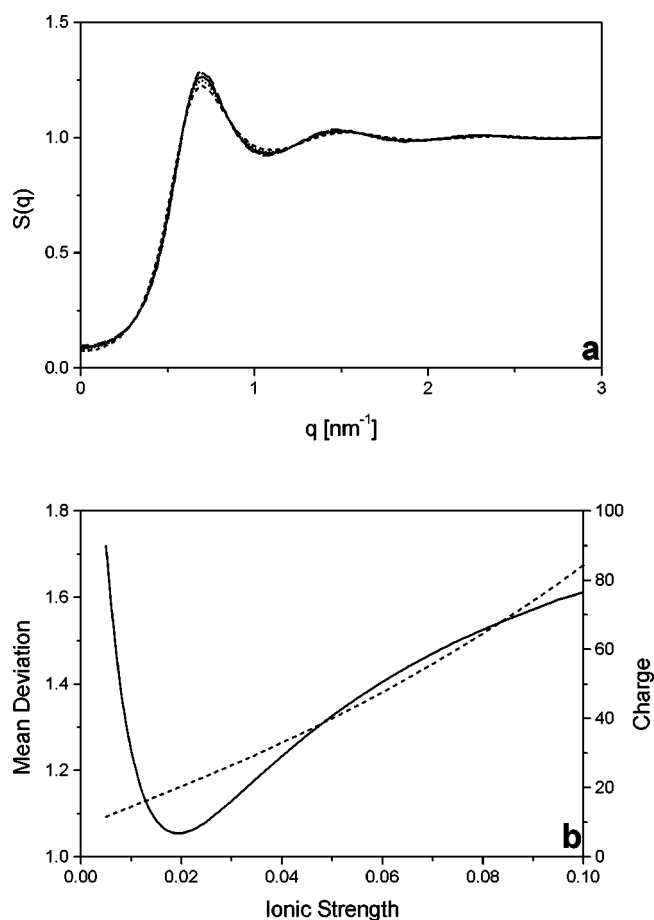


FIG. 3. Simulation like in Fig. 1, variation of the ionic strength: Nearly the same structure factor like in Fig. 1(c) can be found using the RMSA closure relation. (a) $I = 0.01$ (short dashed), $I = 0.015$ (dotted), $I = 0.020$ (solid line), $I = 0.025$ (dashed dotted), $I = 0.030$ (long dashed). (b) Resulting charge as a function of the ionic strength (dashed line) and the corresponding mean deviation of the approximated scattering curve (solid line). The exact ionic strength in the simulation was 0.02.

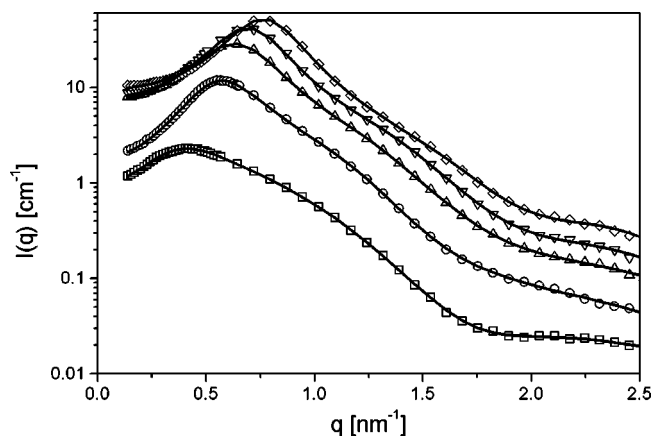


FIG. 4. Experimental SANS data and approximations by the GIFT technique using the HNC closure relation for different CTAB solutions in 10 mM KCl at 37 °C. 1% ($\square$), 5% ($\circ$), 10% ($\triangle$), 15% (∇), and 20% ($\diamond$).

celles is about 4–5 nm while the maximum dimension is at least 14 nm.

C. Different closure relations

The different closure relations for charged particles (5% CTAB in 10 mM KCl) give nearly the same result in terms of PDDFs (Fig. 9). Even the PDDF obtained for a structure factor of uncharged hard spheres is basically the same as a nearly spherical micelle and differs only slightly around the maximum dimension of the particle. The resulting parameters of the structure factor are, at least for the closure relations of charged particles, within the expected range (Fig. 10). The concentration variation gives a nearly linear increase of effective volume fraction with concentration, while the interaction radius increases only slightly according to the small elongation. The charge stays constant over the whole concentration regime studied.

VI. DISCUSSION

The simulations demonstrate that it is possible to calculate the PDDFs of charged interacting particles using the GIFT technique. The resulting PDDFs are very close to the

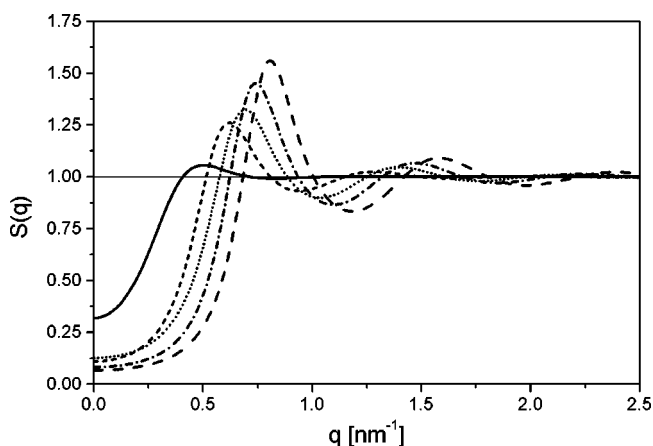


FIG. 5. Structure factors (HNC closure relation) of CTAB solutions in 10 mM KCl at 37 °C. 1% (solid line), 5% (short dashed line), 10% (dotted line), 15% (dash dotted line), and 20% (long dashed line).

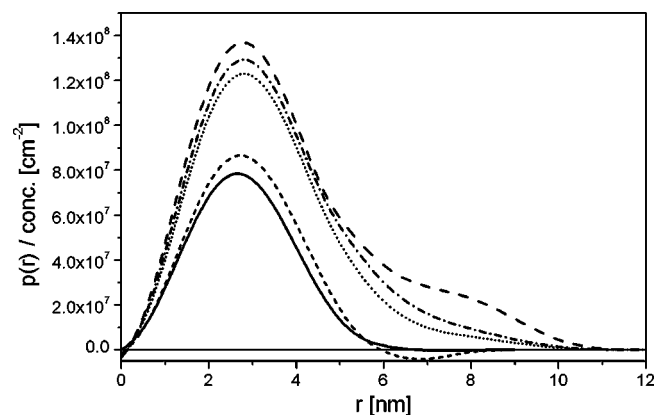


FIG. 6. Pair distance distribution functions of CTAB solutions in 10 mM KCl at 37 °C. Interaction effects have been eliminated by means of structure factors using the HNC closure relation. 1% (solid line), 5% (short dashed line), 10% (dotted line), 15% (dash dotted line), and 20% (long dashed line).

one where the theoretical structure factor used for the simulation has been taken into account and the results obtained by all three closure relations that have been examined are nearly identical [Fig. 1(b)]. Consequently the approximated curves to the simulated scattering intensity are nearly indistinguishable for all three structure factor models [Fig. 1(a)]. All structure factors models can approximate the low- q part and the first peak well [Fig. 1(c)]. The only obvious difference between the theoretical structure factor and the approximated results is for the RMSA for q values higher than 1 nm^{-1} , but even this discrepancy is only about 1% of the theoretical value. The micellar shape, obtained for most of the CTAB samples is nearly spherical, but with clear deviation of this ideal symmetry. This ellipsoidal shape of these micelles is well known.^{23,24} The GIFT technique also finds the transition from nearly spherical to elongated micelles with increasing concentration, which is reported to take place at 0.27 M (corresponding to about 9.8%)²⁴ (Fig. 6). The overall appearance of the PDDFs at these concentrations is not exactly as one would assume for monodisperse cylinders,² but an obvious “tail” at large distances gives some idea of the linear descent, expected for homogeneous rods. But such a linear decrease would be quite astonishing, because the interaction effects are dominating the measured scattering curve. All the

TABLE I. Aggregation number compared to charge of various CTAB solutions. Interaction effects have been eliminated by means of a HNC-structure factors.

CTAB (%)	KCl (mM)	KBr (mM)	$P(0)$ (cm^{-1})	Aggregation number	Charge (e)	$f(e)$
1	10	...	2.94	169	22	0.13
5	10	...	16.18	187	30	0.16
10	10	...	56.11	324	26	0.08
15	10	...	90.18	347	29	0.08
20	10	...	105.83	305	28	0.09
5	30	...	17.06	197	32	0.16
5	80	...	19.92	230	29	0.13
5	...	10	16.20	187	29	0.16
5	...	30	29.19	337	26	0.08
5	...	80	46.49	536	16	0.03

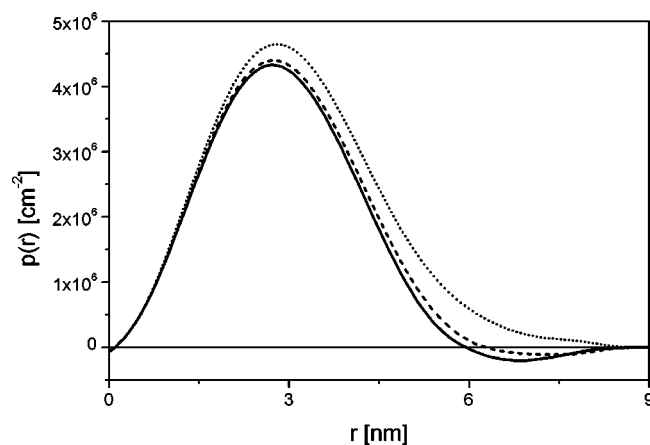


FIG. 7. Pair distance distribution function of 5% CTAB solutions. The interaction potential has been changed by different ionic strengths: 10 mM KCl (solid line), 30 mM KCl (dashed line), and 80 mM KCl (dotted line).

structure factors tried for these calculations are approximations for spherical models. Therefore one has to be lucky to be able to conclude a minimum length of the cylinders present in the sample. The qualitative information, whether spheres or cylinders are forming can be deduced in any case.

Strictly speaking the factorization into form factor and structure factor according to Eq. (2) is only possible for spherical particles due to orientation correlations. But it has been shown for low concentrations that an evaluation by means of the GIFT technique is possible if the length of the particles is on the order of magnitude of the radius.²⁷ This allows one to recover the form factor of the particle qualitatively but needs a certain amount of polydispersity within the structure factor model. The structure factor models used in this contribution for charged particles are for monodisperse spheres and the nonsphericity of the particles is only taken into account in the form factor and its Fourier transforms, the PDDF. This works well for slightly elliptical particles, but the errors made in the PDDF and the structure factor parameters increase with elongation. This is due to the fact that elongation leads to diminished formation of interaction peaks, which can be sometimes partially counterbalanced by the structure factor parameters. But an essential feature of the

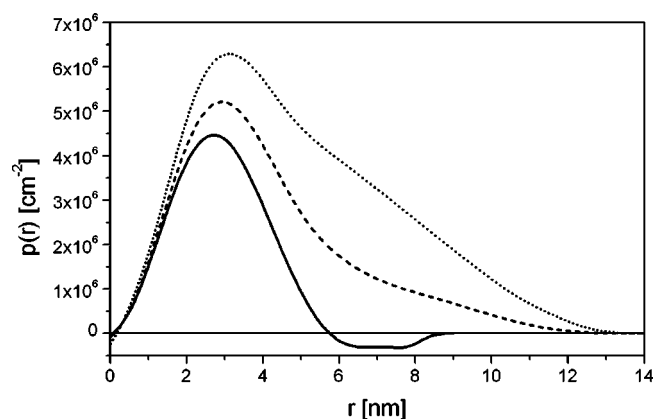


FIG. 8. PDDFs of 5% CTAB solutions. The addition of KBr induces an elongation of the particles. 10 mM KBr (solid line), 30 mM KBr (dashed line), and 80 mM KCl (dotted line).

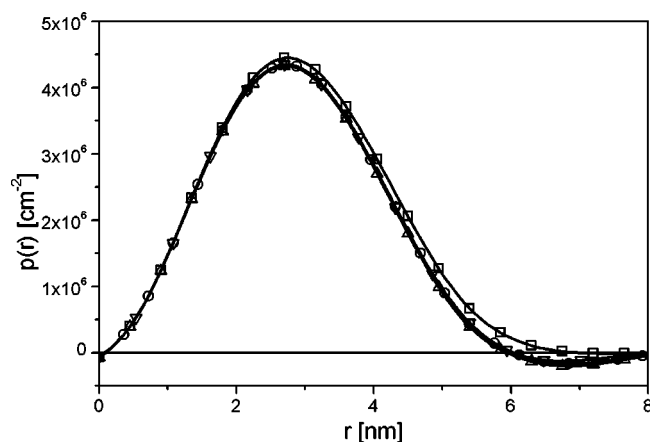


FIG. 9. Comparison of the PDDFs of a 5% CTAB solution in 10 mM KCl. The interaction effects have been taken into account by different structure factor models: Rescaled mean spherical approximation (RMSA, ∇), hypernetted chain (HNC, $\triangle$) and Rogers-Young (RY, $\circ$). The result obtained with a Percus-Yevik (PY) averaged structure factor for uncharged, but polydisperse spheres is also given ($\square$).

structure factors of spherical as well as nonspherical particles are the lowered scattering intensities in the low- q regime, which can be described reasonably by the models for spherical particles, although the structure factor parameters will deviate from the correct ones for cylindrical particles.

In addition to these problems for elongated micelles such systems should form a size distribution with exponentially decaying lengths.²⁸ Such a size distribution would cause PDDFs similar to the ones observed for the CTAB solutions studied.²⁹ Nevertheless it would be difficult to differentiate between a distribution of short cylinders and long cylinders, where the interaction effects could not be eliminated completely.

The addition of KBr leads, contrary to KCl, to a strong elongation of the CTAB micelles.^{25,26} The PDDFs show this development clearly, although the limitations for this type of particle have already been mentioned. But in this case, the overall concentration is lower and the ionic strength is increased. Both factors lead to a weaker interaction compared to the cylinders in the concentration series and the PDDFs correspond better to the expected shape of elongated particles (Fig. 8).

The deviations from the ideal spherical shape of the micelles might also give rise to the small minimum in the PDDF, found close to the maximum dimension for evaluations using the RMSA, HNC, and RY schemes (Fig. 9). The corresponding PDDF calculated with a structure factor for uncharged hard spheres, but with the possibility of some polydispersity, gives, at first glance, a better result. Nevertheless, the parameters for volume fraction and radius, obtained for the uncharged hard spheres, are, as expected, far off the correct values (e.g., a volume fraction of 0.28 for a 5% solution).

The parameters obtained by the GIFT technique for charged particles are the volume fraction, the charge, and the interaction radius. But these parameters show an important linear dependence leading to correct PDDF and $S(q)$ functions, but the values obtained for the $S(q)$ parameters vary

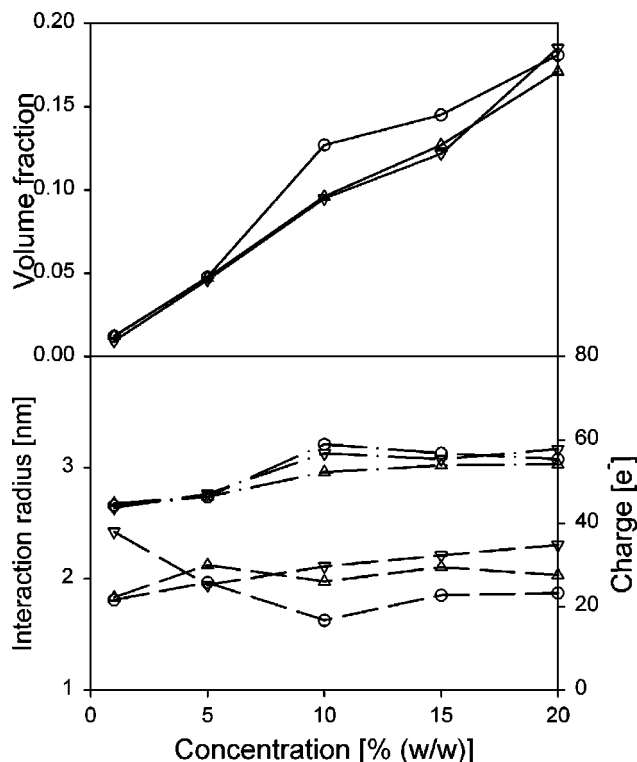


FIG. 10. The values of the parameters used to calculate the structure factors of different CTAB concentrations with various models (RMSA: ∇ , HNC: $\triangle$, and RY: $\circ$). The three parameters calculated are volume fraction (solid lines), charge (dashed lines), and interaction radius (dashed dotted lines).

strongly (Fig. 2). To improve this situation, a second GIFT calculation can be performed. But this time one can use the facts that the structure factor models used are for spherical particles and that the shape of the PDDF is already correct. Deconvolution of the PDDF using spherical symmetry gives a good estimation for the particle radius and herewith for the interaction radius. The radius can be set to this fixed value in the second GIFT run, giving reasonable values for the other structure factor parameters. The values for the volume fraction acquired by this method from our experimental data correlate correctly with the sample concentration. The charges also agree well with the values f (number of charges per macroion divided by the aggregation number) reported in the literature³⁰ (Table I), although the RMSA gives somewhat too high values for higher salt concentrations (results not shown).

The calculation of structure factors of charged particles requires, in any case, several parameters. But the problem is already highly nonlinear for uncharged particles and an even higher number of possible fit parameters would lead to severe numerical instabilities besides the linear dependence. Therefore parameters that can be obtained by other means, such as temperature and dielectric constant of the solvent, have to be treated as fixed input parameters. But with charge and ionic strength this selection is not obvious, although due to numerical limitations only one of the two parameters can be varied. The Yukawa potential, however, used for the calculations is based on a number of assumptions. This causes a rescaling of the charge to an effective charge which has less physical sense.³¹ Consequently it is better to enter the

ionic strength, which can be calculated from the salt concentration, than an effective charge, which cannot be derived directly from the physical charge, as a fixed parameter. Figure 3(a) shows that an error of 50% in the assumed ionic strength can be compensated by an adjusted charge to a very high degree. The resulting parameters are of course wrong, if an incorrect ionic strength has been used as input parameter, but the quality of the fit is not highly effected [Fig. 3(b)]. In particular, if the ionic strength has been assumed too high, the mean deviation is not much worse than for the theoretical solution, leading to small differences in the resulting PDDFs.

The Boltzmann simplex simulated annealing algorithm used in the GIFT technique needs to calculate several thousand different structure factors. This can lead to notable calculation time, especially for very accurate closure relations like the Rogers–Young approximation. But this high accuracy is only necessary in special cases. Most times the results obtained for the hypernetted chain approximations are of equal quality within the limits of the GIFT technique. The rescaled mean spherical approximation shows some minor deviations of the resulting parameters from the expected ones, but it is by far the fastest of the three techniques and at least the resulting PDDFs are nearly the same like for the other more sophisticated closure relations. Therefore RMSA is best for a quick evaluation, while HNC can be used as a standard for GIFT calculations of charged particles. RY is, at least at the moment, quite slow on a PC (up to 12 h), but it might be useful for very detailed calculations.

ACKNOWLEDGMENTS

We want to thank Bruno D' Aguanno for providing us with his program P3 for RY and HNC. We wish to express our thanks to Joachim Kohlbrecher for his help during the SANS experiments. This work was supported by the "Österreichischer Fonds zur Förderung wissenschaftlicher Forschung," Grant No. P12611-CHE.

- ¹A. Guinier and G. Fournet, *Small-Angle Scattering of X-rays* (Wiley, New York, 1955).
- ²*Small-Angle X-ray Scattering*, edited by O. Glatter and O. Kratky (Academic, London, 1982).
- ³L. A. Feigin and G. I. Svergun, in *Structure Analysis by Small-Angle X-ray and Neutron Scattering*, edited by G. W. Taylor (Plenum, New York, 1987).
- ⁴*Neutron, X-Ray and Light Scattering*, edited by P. Lindner and Th. Zemb (North-Holland, Amsterdam, 1991).
- ⁵*Modern Aspects of Small-Angle Scattering*, edited by H. Brumberger (Kluwer, Dordrecht, 1995).
- ⁶J. P. Hansen and I. R. McDonald, *Theory of Simple Liquids* (Academic, London, 1990).
- ⁷B. D'Aguanno and R. Klein, in *Light Scattering. Principles and Development*, edited by W. Brown (University Press, Oxford, 1996).
- ⁸J. Brunner-Popela and O. Glatter, *J. Appl. Crystallogr.* **30**, 431 (1997).
- ⁹O. Glatter, *J. Appl. Crystallogr.* **10**, 415 (1977).
- ¹⁰A. Bergmann, G. Fritz, and O. Glatter, *J. Appl. Crystallogr.* **33**, 1212 (2000).
- ¹¹L. S. Ornstein and F. Zernike, *Proc. Sect. Sci. K. Ned. Akad. Wet.* **17**, 793 (1914).
- ¹²J. B. Hayter and J. Penfold, *Mol. Phys.* **42**, 109 (1981).
- ¹³J. B. Hansen and J. B. Hayter, *Mol. Phys.* **46**, 651 (1982).
- ¹⁴J. K. Percus and G. J. Yevick, *Phys. Rev.* **110**, 1 (1958).
- ¹⁵F. J. Rogers and D. A. Young, *Phys. Rev. A* **30**, 999 (1984).
- ¹⁶J. Chihara, *Prog. Theor. Phys.* **50**, 409 (1973).
- ¹⁷W. G. Madden and S. A. Rice, *J. Chem. Phys.* **72**, 4208 (1980).
- ¹⁸G. Zerah and J. P. Hansen, *J. Chem. Phys.* **84**, 2336 (1986).
- ¹⁹G. Senatore and L. Blum, *J. Phys. Chem.* **89**, 2676 (1985).
- ²⁰B. D'Aguanno and R. Klein, *J. Chem. Soc., Faraday Trans.* **87**, 379 (1991).
- ²¹D. Gazzillo, A. Giacometti, and F. Carsughi, *J. Chem. Phys.* **107**, 10141 (1997).
- ²²P. Ekwall, L. Mandell, and P. Solyom, *J. Colloid Interface Sci.* **35**, 519 (1971).
- ²³S. S. Berr, *J. Phys. Chem.* **91**, 4760 (1987).
- ²⁴P. S. Goyal, R. Chakravarthy, B. A. Dasannacharya, J. A. E. Desa, V. K. Kelkar, C. Manohar, S. L. Narasimhan, K. R. Rau, and B. S. Valaulikar, *Physica B* **156–157**, 471 (1989).
- ²⁵V. Rajagopalan, P. S. Goyal, B. S. Valaulikar, and B. A. Dasannacharya, *Physica B* **180–181**, 525 (1992).
- ²⁶P. S. Goyal, S. V. G. Menon, B. A. Dasannacharya, and V. Rajagopalan, *Chem. Phys. Lett.* **211**, 559 (1993).
- ²⁷B. Weyerich, J. Brunner-Popela, and O. Glatter, *J. Appl. Crystallogr.* **32**, 197 (1999).
- ²⁸M. E. Cates and S. J. Candau, *J. Phys.: Condens. Matter* **2**, 6869 (1990).
- ²⁹J. Brunner-Popela, R. Mittelbach, R. Strey, K.-V. Schubert, E. W. Kaler, and O. Glatter, *J. Chem. Phys.* **110**, 10623 (1999).
- ³⁰P. S. Goyal, B. A. Dasannacharya, V. K. Kelkar, C. Manohar, R. Srinivasa Rao, and B. S. Valaulikar, *Physica B* **174**, 196 (1991).
- ³¹L. Belloni, *J. Chem. Phys.* **85**, 519 (1986).